WASP-2b

R-1458

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_220912` · created 2026-08-01T10:28:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459834.7862 +/- 0.007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.046
Transit depth (Rp/Rs)^2	2.56 +/- 1.46 [%]
Orbital Inclination (inc)	88.26 +/- 2.84
Airmass coefficient 1 (a1)	0.54 +/- 0.26
Airmass coefficient 2 (a2)	0.35 +/- 0.14
Scatter in the residuals of the lightcurve fit is	41.02 %
Best Comparison Star	None
Optimal Aperture	15.81
Optimal Annulus	20.02
Transit Duration (day)	0.082 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.046 vs 0.1326 (deviation 0.43σ)
Duration fit vs published	0.082 d vs 0.0737 d (deviation 0.4σ)
Mid-time O–C	-17.0 min
Depth SNR	2.5
Residual scatter	41.02 %

Light curve

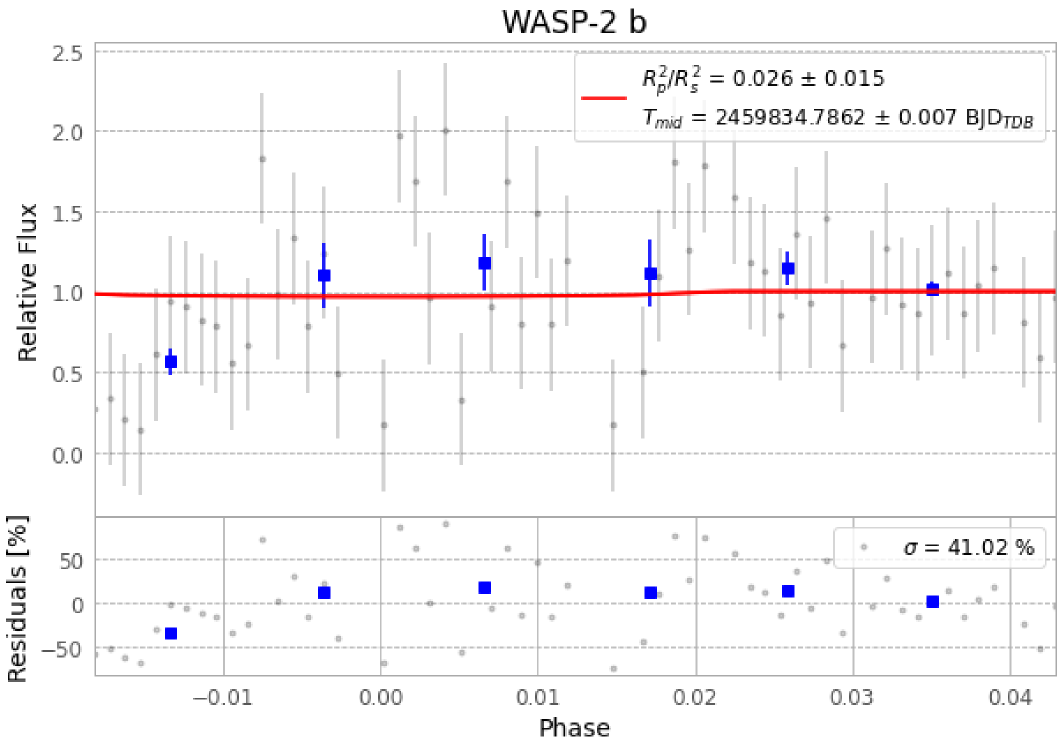

Figure 1 — FinalLightCurve_WASP-2 b_2022-09-11.png (SHA-256 c43b3c248584a019…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [263, 459], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 04f21ea7e34ebcde…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

63 FITS frames (41 MB), `WASP-2220912054813.FITS` → `WASP-2220912085712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-220912.log (cc306a6185c0…), FinalLightCurve_WASP-2 b_2022-09-11.png (c43b3c248584…), FinalLightCurve_WASP-2 b_2022-09-11.csv (75d43d835d49…)

Compute resources

Wall time 250.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-01 10:28Z.